FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014: 2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA INTEGRATED TEST SERIES

FIITJEE

JEE(Advanced), 2015

ANSWERS, HINTS & SOLUTIONS

HALF COURSE TEST –VII
(Paper-1)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	B	A	A
2.	D	C	B
3.	A	C	C
4.	B	C	A
5.	D	A	C
6.	C	B	B
7.	A, C	B, C, D	A, B, C
8.	B, C	A, B, C, D	A, C
9.	B, D	A, B, C, D	A, C, D
10.	B, D	A, B, D	A, D
11.	D	B	D
12.	D	C	D
13.	D	D	A
14.	A	A	B
15.	C	B	C
16.	C	B	A
17.	A	C	B
18.	A	A	B
19.	B	A	C
20.	D	D	A
21.	C	B	B
22.	A	A	C
23.	D	C	B

Physics

PART – I

1. Power = $\frac{d(K.E.)}{dt}$

2. The force of friction, when it is not slipping, is less than its limiting value : $|f| \le \mu N$. This leads to the condition on F.

3. Velocity of upper block when lower block hits a obstacle,

$v = \sqrt{2a.\ell} = \sqrt{2.\frac{F}{2M}\ell} = \sqrt{\frac{F\ell}{M}}$

now after collision retardation of upper block w.r.t. earth,

$a = \frac{\mu Mg}{M} = \mu g$

$\therefore \quad v^2 = u^2 + 2as$

$\therefore \quad 0 = \frac{F\ell}{M} - 2\mu g.\frac{\ell}{2}$

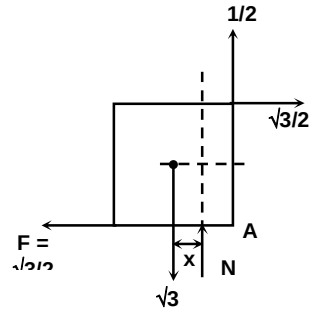
$\Rightarrow \quad \mu = \frac{F}{Mg}$.

4. For equilibrium

$\Sigma\vec{F} = 0 \;\; \Sigma\vec{\tau} = 0$

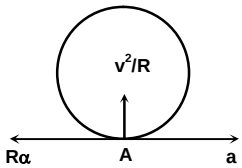
$N = \frac{\sqrt{3}}{2} - \frac{1}{2}$ ….(i)

Torque about A due to all forces excluding normal force is found to be zero.

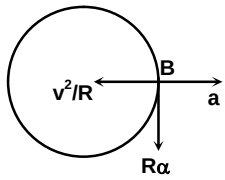


5. Impulse = Force × time; net momentum of the system is conserved as the net force on it is zero.

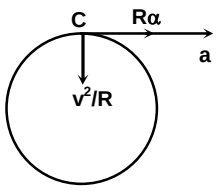
6. At A :
for a rolling wheel, $a = R\alpha$
$\therefore$ (A) is correct.

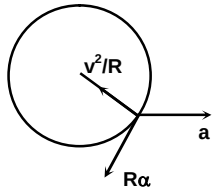


At B :
If $\frac{V^2}{R} = a$ then a_B may be vertically downwards
$\therefore$ (B) is correct.
Consider this
$\therefore$ (D) is correct.



At C :
$\therefore$ (C) is correct.





15. $R_{eff} = \int_{R}^{2R} \frac{dx}{K(4\pi x^2)} + \int_{2R}^{3R} \frac{dx}{2K(4\pi x^2)} = \frac{1}{4\pi K}\left[-\frac{1}{x}\right]_{R}^{2R} + \frac{1}{8\pi K}\left[-\frac{1}{x}\right]_{2R}^{3R}$

$\Rightarrow R_{eff} = \frac{7}{48\pi KR}$

17. Consider shell with cross sectional thickness equal to AB, BC and CA respectively. Since heat flow is same through all of them, therefore temperature difference will be maximum across the shell with maximum resistance which will be AB.

18. As explained in figure the no of maximums are 7.

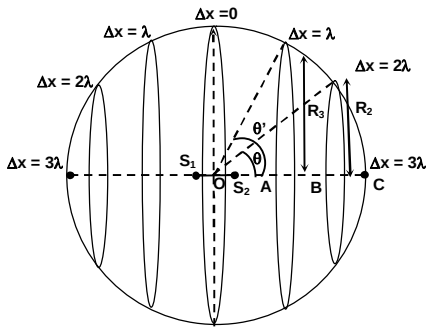
19. $R_2 = R\sin\theta$ and $(3\lambda \cos\theta = 2\lambda) = R\sqrt{5}/3$

Length = $\frac{2\pi R\sqrt{5}}{3}$

20. $OC - OB = R - R\cos\theta = R/3$

$OB - OA = 2R./3 - R\cos\theta' = R/3$ $(3\lambda \cos\theta' = \lambda \cos\theta' = 1/3)$

Hence $R_1 = R\sin\theta' = R\sqrt{8}/3$



21. $N = mg$ $\qquad f = \mu mg$

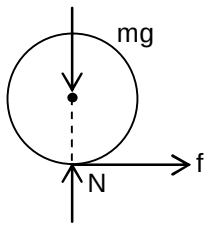
$V = \frac{f}{m}t = \mu gt$

$fR = \mu mgR = \frac{2}{5}mR^2\alpha$

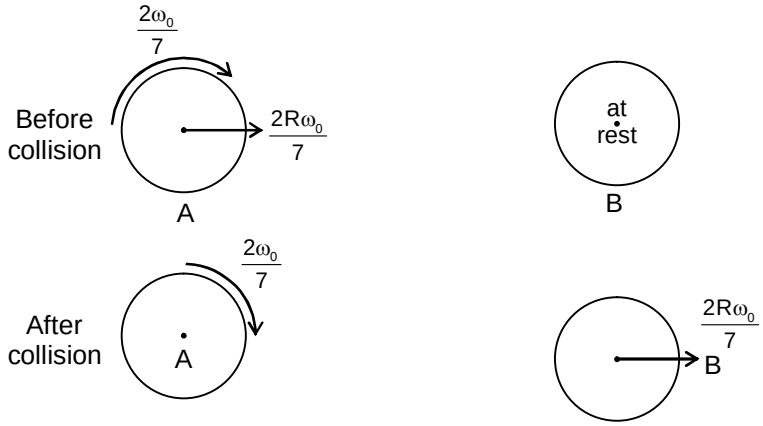
$\Rightarrow \alpha = \frac{5}{2}\mu\frac{g}{R} \quad \omega = \omega_0 - \frac{5\mu g}{2R}t$

$\mu gt = R\omega_0 - \frac{5}{2}\mu gt$

$\Rightarrow t = \frac{2R\omega_0}{7\mu g}$



22.



$$K.E. = \frac{1}{2}m\left[\frac{2R\omega_0}{7}\right]^2 = 2\frac{mR^2\omega_0^2}{49}$$

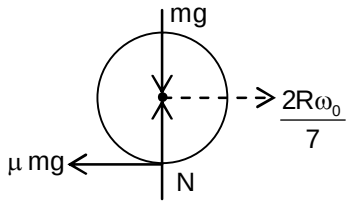
23. $\mu mgR = \frac{2}{5}mR^2\alpha \quad \Rightarrow \quad \alpha = \frac{5}{2}\frac{\mu g}{R}$

$R\omega = \frac{5}{2}\mu gt \qquad V = \frac{2R\omega_0}{7} - \mu gt = \frac{5}{2}\mu gt$

$\Rightarrow \quad \frac{2R\omega_0}{7} = \frac{7}{2}\mu gt \quad \Rightarrow \quad t = \frac{4}{49}.\frac{R\omega_0}{\mu g}$

$\omega = \frac{5}{2}.\frac{\mu g}{R}.\frac{4}{49}\frac{R\omega_0}{\mu g} = \frac{10}{49}.\omega_0$

$V = \frac{10}{49}R\omega_0 \qquad K.E. = \frac{1}{2}I\omega^2 + \frac{1}{2}mV^2$

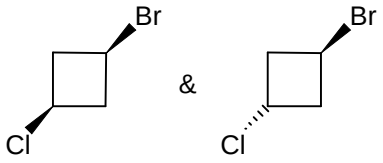


Chemistry

PART – II

1. $\frac{-Ea}{RT} = -\frac{E'_a}{RT'}, \frac{100}{500} = \frac{80}{T'}$, $\therefore$ T′ = 400 K
$\therefore$ Option (A) is the correct choice.

2. Due to conjugation & H – bonding.

3. 

cis trans

4. Less substituted β - H is removed first.

5. PE of a free electron is zero at infinity i.e. out of the field of nucleus.

9. All are true except (A). D_2 also show spin isomerism like H_2.
Hence B, C, D is correct.

10. All compounds A, B, C, D contains chiral centres and hence optically active i.e. chiral.
Hence A, B, C, D is correct.

11. Intensive properties are independent of quantity of substance.
Hence A, B, C, D is correct.

12. A & B are acid buffers while (D) is a basic buffer. (C) is not a buffer because no base (NH_4OH) will be left after the reaction.
Hence A, B, D is correct.

Mathematics PART – III

1. $h(x) = \ln(f(x)\cdot g(x)) = \ln e^{[y]+\{y\}} = \{y\} + [y] = y = e^{|x|}\operatorname{sgn} x$

$$\therefore\ h(x) = e^{|x|}\operatorname{sgn} x = \begin{cases} e^{x} & \text{if } x > 0 \\ 0 & \text{if } x = 0 \\ -e^{-x} & \text{if } x < 0 \end{cases}$$

$$h(-x) = \begin{cases} e^{-x} & \text{if } x < 0 \\ 0 & \text{if } x = 0 \\ -e^{x} & \text{if } x > 0 \end{cases}$$

$h(x) + h(-x) = 0$ for all x

2. For $\arg\left(\frac{a-b}{a-c}\right) = 0 \Rightarrow \frac{a-b}{a-c} > 0$

Select 3 numbers from the set and call the greatest as 'a' and remaining two as b and c. This can be done in ${}^{2013}C_3 \times 2$ ways. Therefore the probability shall be $\frac{{}^{2013}C_3 \times 4}{{}^{2013}C_3 \times 3!} = \frac{2}{3}$

3. $f(x) = ax^2 + bx + c = 0$
Now as $a + b + c < 0 \Rightarrow f(1) < 0$
$a - b + c < 0 \Rightarrow f(-1) < 0$
$c > 0 \Rightarrow f(0) > 0$
Combining (i), (ii) and (iii) to get $f(-1)\, f(0) < 0$ and $f(0)\, f(1) < 0$
Hence one of the root lies between (–1, 0) and other root lies between (0, 1)
$[\alpha] + [\beta] = -1 + 0 = -1$

4. Line is satisfied by infinite points having x coordinate 1 and y coordinate $\in$ R
$(a - b)x + (b - c)y + (c - a) = 0$ ….(1)
$\because$ (1) is satisfied by (1, 1)
Also as (a – b), (b – c), (c – a) are in AP

$$\therefore\ b - c = \frac{(a-b)+(c-a)}{2}$$

Putting in (1)

$$(a-b)x + \left[\frac{(a-b)+(c-a)}{2}\right]y + (c-a) = 0$$

$\Rightarrow (a - b)(2x + y) + (c - a)(y + 2) = 0$
Which represents family of line passing through (1, –2)
$\therefore$ Line (1) will pass through (1, 2) where $y \in R$

5. $x \le 0$ and $y > 0$. Hence $x < y$

6. $3^{2-\log_3 4} = 3^2.(3^{-\log_3 4}) = 9 \times 4^{-1} = \frac{9}{4}$

7. $y = x^{\log_x e} = e$ (constant) $\Rightarrow$ A and C
as $\lim\limits_{x\to 1} f(x) = e$

9. $\frac{2}{1!13!}+\frac{2}{3!11!}+\frac{2}{5!9!}+\frac{1}{7!7!}=\frac{1}{14!}(2^{14}C_1+2^{14}C_3+2^{14}C_5+{}^{14}C_7)=\frac{1}{14!}2^{14-1}=\frac{2^{13}}{14!}$
$\Rightarrow$ m = 13, n = 14

11. By a general Venn-diagram we can conclude that statement is not always true

12. Statement-1 is wrong as z can be written as $\frac{-(x+y)}{1-xy}$
$\Rightarrow$ For any values of x and y; $(xy \neq 1)$ we get a value of z
And Statement-2 is correct
Hence, code (D) is the correct answer

13. Statement-2 is correct as ellipse is a central conic
And it also explains Statement-1
Hence, code (A) is the correct answer

18. We can pile up these 12 books in a single lot in 12! Ways

19. If we pickup the book always from the top of respective lot then these books will behave as alike books in a single lot of those 12 books. It means we have to arrange 12 distinct books in which 3 are alike, 4 are alike and 5 are alike
Hence required way is $\frac{12!}{3!\times 4!\times 5!}$

20. If we arrange Chemistry and Maths book, we can arrange in 8! Ways, and in remaining 9 slots we can place 4 Physics books $8!\times {}^{9}P_4$